

Increasing the Alignment Tolerance in Long-Reach FSO Links by using a Photodiode Matrix

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Abstract A matrix of 2D-interconnected PDs enables a broadband FSO receiver with large effective aperture and wide field-of-view. Without needing adaptive optics, it offers superior link robustness over single-PD and MMF-coupled FSO receivers, and compact receiver solution. Alignment tolerances increase $>3\times$ and scale with matrix size. ©2026 The Author(s)

Introduction

Optical wireless communication using accurately directed narrow beams enables high-capacity links with low latency, high privacy and high EMI immunity, and offers relief to the congestion issues in present RF-based wireless networks.

In short-reach indoor applications, such narrow-beam systems enable many users to be individually connected at high bandwidth [1][2][3], outperforming wide-beam MAC-shared LiFi systems [4]. The confinement of optical beams enables the creation of pico-cells in dense user settings and avoids crosstalk among them.

In long-reach free-space optical (FSO) systems, such as aircraft-to-satellite, ground-to-satellite and ground-to-aircraft links, maximizing the received power requires accurately pointed well-focused beams and large-aperture receivers. The confined beams enable optimized signal power transfer and obstruct link jamming. Only a single beam (or few beams) has to be steered over a narrow angular range with high accuracy. However, this comes at the cost of extremely stringent alignment requirements. State-of-the-art solutions rely on complex and costly adaptive optics and beam tracking systems to couple light into a single photodiode or a fiber (see e.g. [5]), which makes such receivers inherently complex and expensive, and require complex measures for maintaining a stable FSO link to a moving terminal (such as ground-to-aircraft link, cf. [6]).

In prior work [2][7][8][9], we introduced a photodiode (PD) matrix receiver concept for short-reach OWC, primarily targeting wide field-of-view (FoV) applications. In this paper, we show that this concept fundamentally changes the design

trade-offs for long-reach FSO systems. Instead of relying on precise optical coupling, the PD matrix provides a much larger effective aperture, significantly relaxing the alignment constraints.

The key novelty of this work is the demonstration that a PD matrix receiver can eliminate the need for complex adaptive alignment systems in long-reach FSO links, enabling a simpler, more robust, and cost-effective receiver architecture with increased ability to handle fast moving objects. This challenges the conventional paradigm of high-precision alignment in long-distance optical communication systems.

Optical wireless receiver using a matrix of photodiodes

In order to create a broadband receiver with large aperture and wide FoV, we have introduced the photodiode matrix (PD matrix) concept shown in Fig. 1 [7][8][9]. Typically, when enlarging the active area of a photodiode its junction capacitance grows proportionally, and hence the receiver's bandwidth decreases. Therefore, we choose to arrange and interconnect many photodiodes into a 2D square matrix. By consecutive Norton-Thevenin transforms it can be shown that the resulting capacitance of the matrix equals that of a single photodiode, hence the bandwidth is not affected while the active photodetection area is considerably enlarged. The shunt resistors R_p reduce the bias voltage variations at each row of photodiodes which occur when the matrix is unevenly illuminated. We designed a 4×4 matrix of $\varnothing 150\mu\text{m}$ 1.25GHz PIN photodiodes, thus extending the active area to $\varnothing 859\mu\text{m}$. It is preceded by

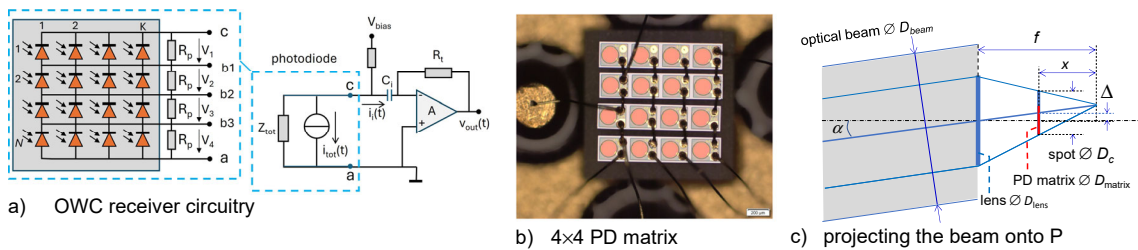


Fig. 1 Broadband OWC receiver with photodiode matrix and lensed input

a $\varnothing 50$ mm $f=10$ mm Fresnel lens, and yields a receiver with $\varnothing 50$ mm aperture and FoV $\approx \pm 10$ deg. Followed by a single 750MHz TIA, we showed 1.25Gbit/s bi-directional OWC indoor links [10]. Adopting this concept, a sectorized 8x8 matrix of $\varnothing 120\mu\text{m}$ photodiodes was reported with an active area of $\varnothing 2.0$ mm and 2.5GHz bandwidth [11].

Other receiver solutions increasing the light capturing aperture were reported in literature, such as a compound parabolic concentrator (CPC) [12], a fluorescent slab concentrator (breaking the law of etendue) [13][14], a single surface grating coupler (SGC) or array of SGCs [15], and an angular diversity receiver (ADR) employing multiple photodiodes each followed by a pre-amplifier [16][17][18]. An ADR based on an array of 12 avalanche photodiodes (APDs) can further improve the receiver sensitivity [19].

Our photodiode matrix solution offers a compact and simple yet broadband receiver architecture; it only needs a single pre-amplifying TIA, consumes minimal energy, has good SNR performance (better than an ADR [7]), and is well suited for monolithic integration.

Impact on receiver performance of the beam spot size incident on the PD matrix

Fig. 1.c shows the coupling of the incoming beam with power P_{beam} and diameter D_{beam} to an $N \times N$ PD matrix with diameter D_{matrix} by means of a lens

with diameter D_{lens} and focal length f . Typically, $D_{beam} > D_{lens}$, and the diameter of the spot incident on the matrix $D_c = p D_{lens}$. When assuming an ideal aberration-free lens and a uniform beam which evenly illuminates the matrix, and when putting the matrix out of the lens' focus by a defocusing factor $p = x/f$ (with $0 \leq p < 1$), the factor T_{eq} with which the beam power is coupled to the matrix and the maximum half-angle α_{max} at which the beam is accepted (i.e. the FoV) are given by [7][9]

$$T_{eq} = \frac{\eta}{N} \left(\frac{D_{matrix}}{p D_{beam}} \right)^2 \cos \alpha \quad \text{for } D_{matrix}/D_{lens} < p < 1$$

$$T_{eq} = \frac{\eta}{N} \left(\frac{D_{lens}}{D_{beam}} \right)^2 \cos \alpha \quad \text{for } 0 \leq p \leq D_{matrix}/D_{lens}$$

$$\tan \alpha_{max} = \frac{|p D_{lens} - D_{matrix}|}{2 f (1 - p)}$$

where η is the fill factor of the matrix. The photodiode current i_{ph} generated by the PD matrix is $i_{ph} = T_{eq} \mathcal{R} P_{beam}$, where $\mathcal{R}$ is the photodiode responsivity. For a non-uniform beam (e.g., with Gaussian profile) which illuminates the matrix unevenly, we find [9]

$$T_{eq} = \frac{1}{N} \sum_{m=1}^N \sum_{k=1}^N a_{mk} = N \overline{a_{mk}}$$

where $\overline{a_{mk}}$ is the average fraction of P_{beam} coupled into photodiode (m,k) .

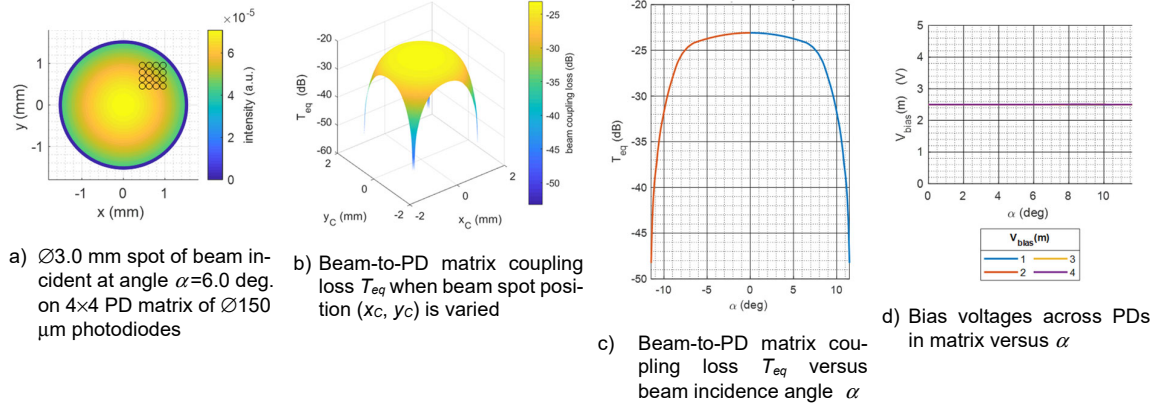


Fig. 2 Performance of OWC receiver for a large beam spot ($\varnothing 3.0$ mm, amply overlapping the 4×4 PD matrix)

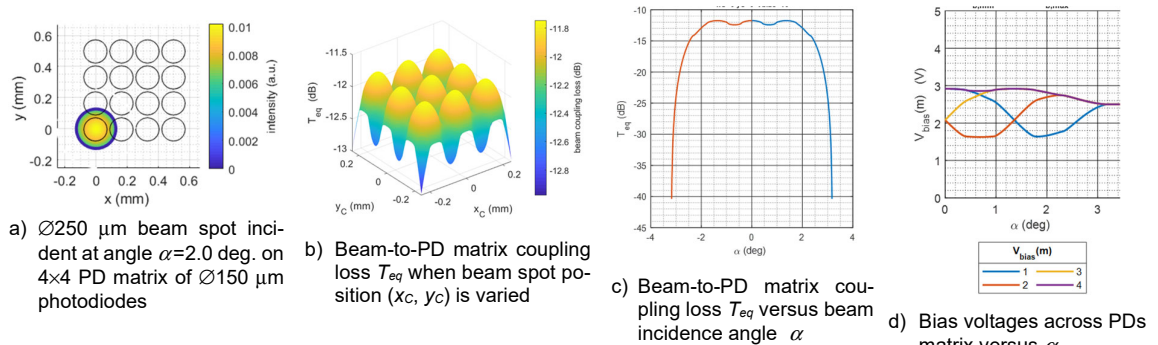


Fig. 3 Performance of OWC receiver for a narrow beam spot ($\varnothing 250 \mu\text{m}$, overlapping a single PD)

In *short-reach OWC links* (such as for indoor applications) typically a large FoV is needed. The FoV of the PD matrix receiver is increased when p increases, i.e. by moving the matrix closer to the lens and thus letting the beam's spot overfill the matrix. This, however, also reduces the beam-to-matrix power coupling factor T_{eq} . Hence, a compromise between the desired FoV and required T_{eq} needs to be set. Fig. 2 shows the receiver performance when the beam spot diameter is $\varnothing 3.0$ mm, i.e. much larger than the $\varnothing 859 \mu\text{m}$ diameter of our 4×4 matrix of $\varnothing 150 \mu\text{m}$ photodiodes in order to get a large FoV. When assuming a +10 dBm Gaussian beam with $\varnothing 100$ mm waist diameter and a $\varnothing 50$ mm aberration-free lens with focal length $f=10$ mm, Fig. 2.a shows how the beam spot arrives on the matrix at beam incidence angle $\alpha=6$ deg., and Fig. 2.b how the coupling loss T_{eq} varies when α is varied and thus the spot center (x_c, y_c) is moved with respect to the center of the matrix. Fig. 2.c shows that a $T_{eq, max}=-23.1$ dB and FoV $\alpha_{max} \approx \pm 9.6$ deg. is achieved (α_{max} is measured at $T_{eq}=-30$ dB, i.e. meeting a -20 dBm receiver sensitivity). When the beam spot moves across the PD matrix, and the total bias voltage across the column of 4 photodiodes is set at 10 V, Fig. 2.d shows that bias voltages per photodiode vary around 2.5 V by only ± 9 mV, due to the shunt resistors $R_p=25\text{k}\Omega$.

In *long-reach OWC links* (such as terrestrial links between buildings, or ground-to-satellite links), such a large FoV is typically not required, but a larger power coupling factor T_{eq} is preferred. Hence, the beam spot size may be reduced. Fig. 3 shows the performance of the 4×4 PD matrix receiver for a beam spot of only $\varnothing 250 \mu\text{m}$. With the same system assumptions as before, Fig. 3.a shows the beam spot incident at $\alpha=2$ deg. on the matrix, and Fig. 3.b how T_{eq} varies when moving the beam's center (x_c, y_c) over the matrix (so varying α). A $T_{eq, max}=-11.7$ dB and FoV $\alpha_{max} \approx \pm 3.1$ deg. is achieved (see Fig. 3.c). When moving the beam spot across the PD matrix, the bias voltages (using the same R_p) across the photodiodes vary a bit more than with the $\varnothing 3.0$ mm beam spot, i.e. by ± 0.6 V around 2.5 V, as Fig. 3.d shows.

Summarizing, shown in Fig. 4, a single-PD receiver has a lower beam-to-photodetector loss T_{eq} than the PD matrix receiver at small spot diameters, but a higher loss at larger spot sizes. The FoV of the PD matrix receiver exceeds that of the single-PD receiver at all spot sizes; notably, at small spot sizes its FoV (and thus the alignment tolerance) is $2.3\times$ larger at the same $T_{eq, max}$. A $50/125\mu\text{m}$ fiber-coupled PD clearly performs worse at all spot sizes. Hence, for long-reach OWC links the PD matrix receiver allows the least

sophisticated optical imaging systems (and thus a larger spot size), and can be more cost-efficient. It thus can notably outperform a single-PD receiver and an MMF-coupled one.

The FoV and beam coupling loss of a PD matrix receiver further improve when increasing the matrix size. Fig. 5 shows how upscaling enlarges the FoV and thus the alignment tolerance by $>3\times$ given a typically required beam-to-matrix power coupling factor $T_{eq}=-30$ dB. The feasibility of upscaling was demonstrated by a receiver with an integrated 20×20 matrix of $\varnothing 30 \mu\text{m}$ photodiodes in a NRZ 25 Gbit/s link of 20 meters [20].

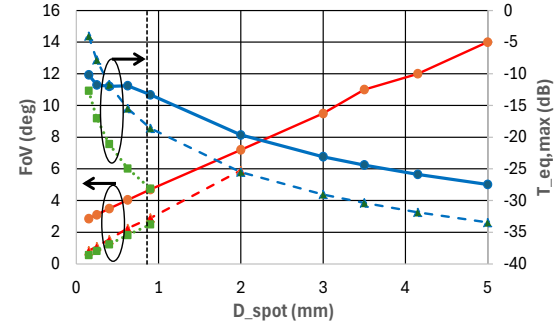


Fig. 4 Performance of 4×4 photodiode matrix-based receiver (solid graphs) versus receiver with single PD (dashed) and MMF-coupled PD (dotted); vertical dotted line: diameter of PD matrix; FoV is measured at $T_{eq} = -30$ dB with $\varnothing 50$ mm lens and $\varnothing 100$ mm beam)

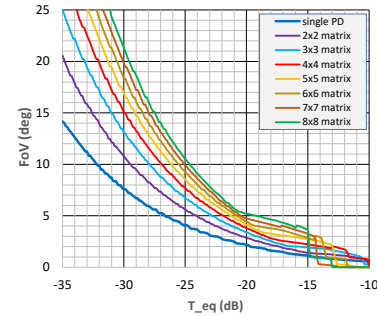


Fig. 5 Performance if the photodiode matrix is scaled up

Concluding remarks

An OWC receiver based on a PD matrix does not need precise optical alignment. It combines simplicity, low power consumption, and better optical performance regarding beam coupling loss and Field-of-View. While initially developed for short-reach applications, it challenges the conventional paradigm in long-reach FSO systems, where complex adaptive optics and fiber-coupled receivers are considered essential. A PD matrix can offer a $>3\times$ increased alignment tolerance with respect to a single PD at the same coupling loss, and clearly outperforms an MMF-coupled PD. A PD matrix enables a simpler, more robust and cost-effective receiver, is readily scalable to further improve performance, avoids complex adaptive optics, and eases a stable FSO link to a moving terminal (such as an aircraft).

Acknowledgments

Acknowledgments are due to the European Research Council for partly funding this research in the Advanced Investigator Grant project BROWSE and Proof-of-Concept grant BROWSE+, as well as to the Dutch Research Council (NWO) in the Gravitation program Integrated Nanophotonics. Wim Korevaar of TNO Optical Satellite Communications and Niek Doelman of TNO Opto-Mechatronics are gratefully acknowledged for fruitful discussions.

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